

Product Carbon Footprint Study Report

Product Name: diammonium phosphate

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: diammonium phosphate

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg diammonium phosphate

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg diammonium phosphate.

2. System Boundary

diammonium phosphate production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of diammonium phosphate produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.982 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

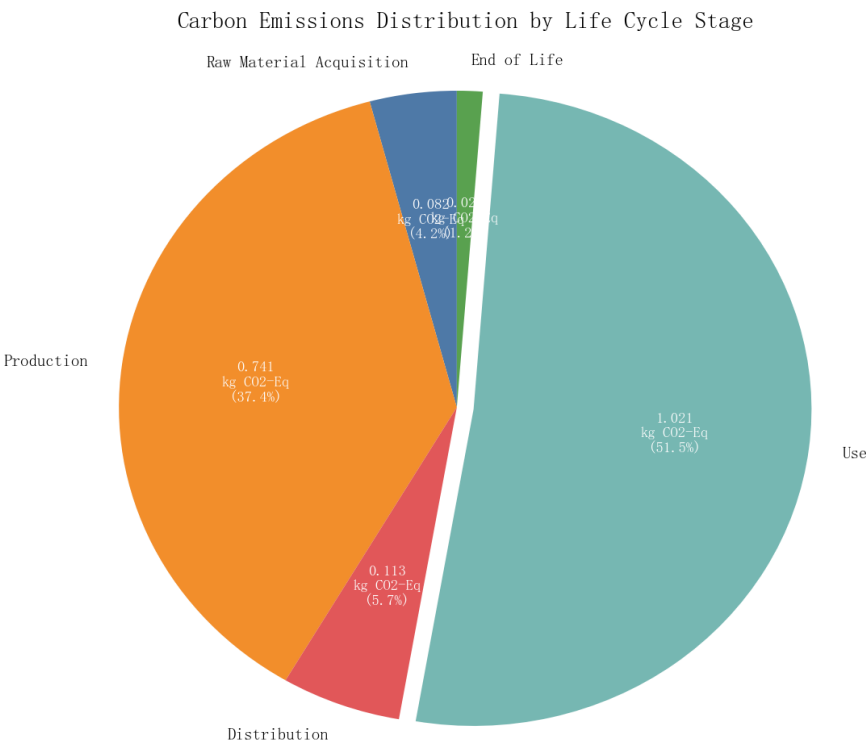


Table 1: Carbon Emissions by Life Cycle Stage

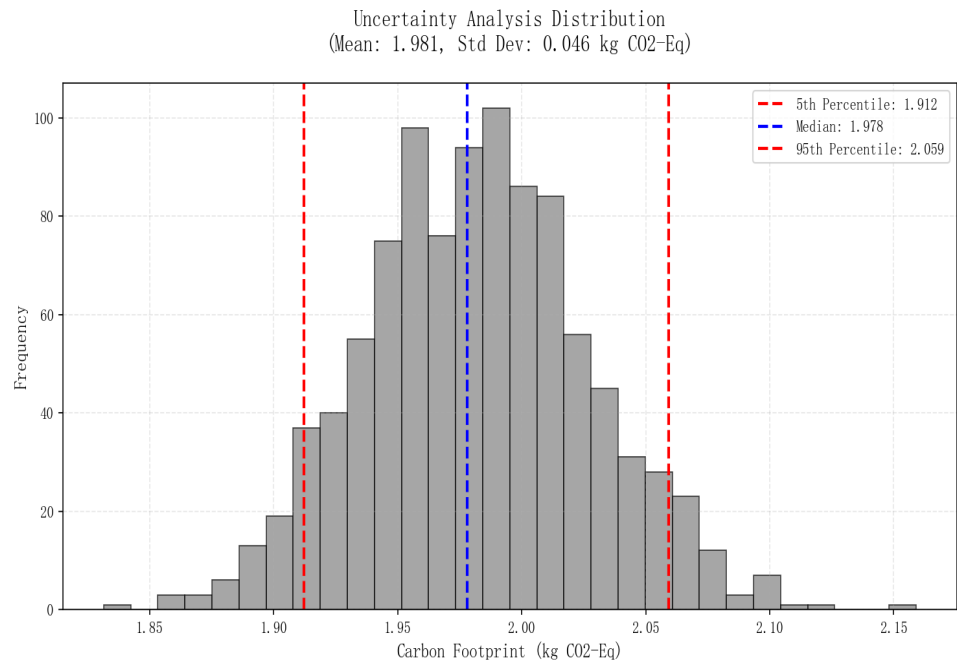
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.082	4.15
Production	0.741	37.39
Distribution	0.113	5.72

Use	1.021	51.52
End of Life	0.024	1.23
Total	1.982	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1.981
Standard Deviation	0.046
Coefficient of Variation (CV)	2.33%
Median	1.978
5th Percentile	1.912
95th Percentile	2.059

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of diammonium phosphate stands at 1.982 kg CO₂-Eq, representing a moderate impact level typical for nitrogen-based fertilizers when assessed on a per-unit basis. This value underscores the product's environmental profile within agricultural supply chains, where emissions arise predominantly from energy-intensive processes and field applications rather than material sourcing or disposal.

Lifecycle stage analysis shows pronounced concentration in two phases: the use stage accounts for 51.5% (1.021 kg CO₂-Eq), while production contributes 37.4% (0.741 kg CO₂-Eq). In contrast, raw material extraction (4.2%), distribution (5.7%), and end-of-life treatment (1.2%) remain marginal, indicating that upstream and downstream logistics exert limited influence relative to manufacturing and application activities.

Major emission sources are therefore identified in the use phase, likely linked to soil-related greenhouse gas releases during fertilizer application, and in production, driven by ammonia synthesis and energy consumption in chemical conversion processes.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a functional unit of one kilogram of diammonium phosphate delivered to the end user, average European transport distances for distribution modeling, and standard agricultural application rates without site-specific variability. These assumptions enable a streamlined cradle-to-grave evaluation while simplifying complex real-world conditions.

Limitations of data sources and quality stem from reliance on generic industry databases such as Ecoinvent for production inventories, which may not fully capture proprietary process efficiencies or regional electricity grid mixes. Data gaps in use-phase emissions, particularly variable soil nitrous oxide factors, introduce uncertainty that could affect result precision by up to 20%.

System boundary restrictions encompass a full lifecycle from raw material extraction through end-of-life disposal but exclude indirect land-use changes, capital equipment depreciation, and downstream food system effects, thereby focusing the assessment on direct product-related emissions only.

3. Suggestions for Improvement

- Optimize fertilizer application rates and timing through precision agriculture technologies to reduce use-phase emissions, targeting a potential 15-25% cut in the 1.021 kg CO₂-Eq hotspot.
- Implement energy-efficient ammonia synthesis and heat recovery systems in production facilities to lower the 0.741 kg CO₂-Eq contribution by adopting low-carbon hydrogen sources.
- Shift to renewable electricity for manufacturing processes and localized sourcing of raw materials to decrease overall footprint while maintaining product functionality.

- Enhance data quality by integrating site-specific primary data from producers and field trials, supplemented by regular updates to emission factors from verified databases.
- Explore alternative end-of-life scenarios such as nutrient recovery to further minimize the already low 0.024 kg CO₂-Eq disposal impact.